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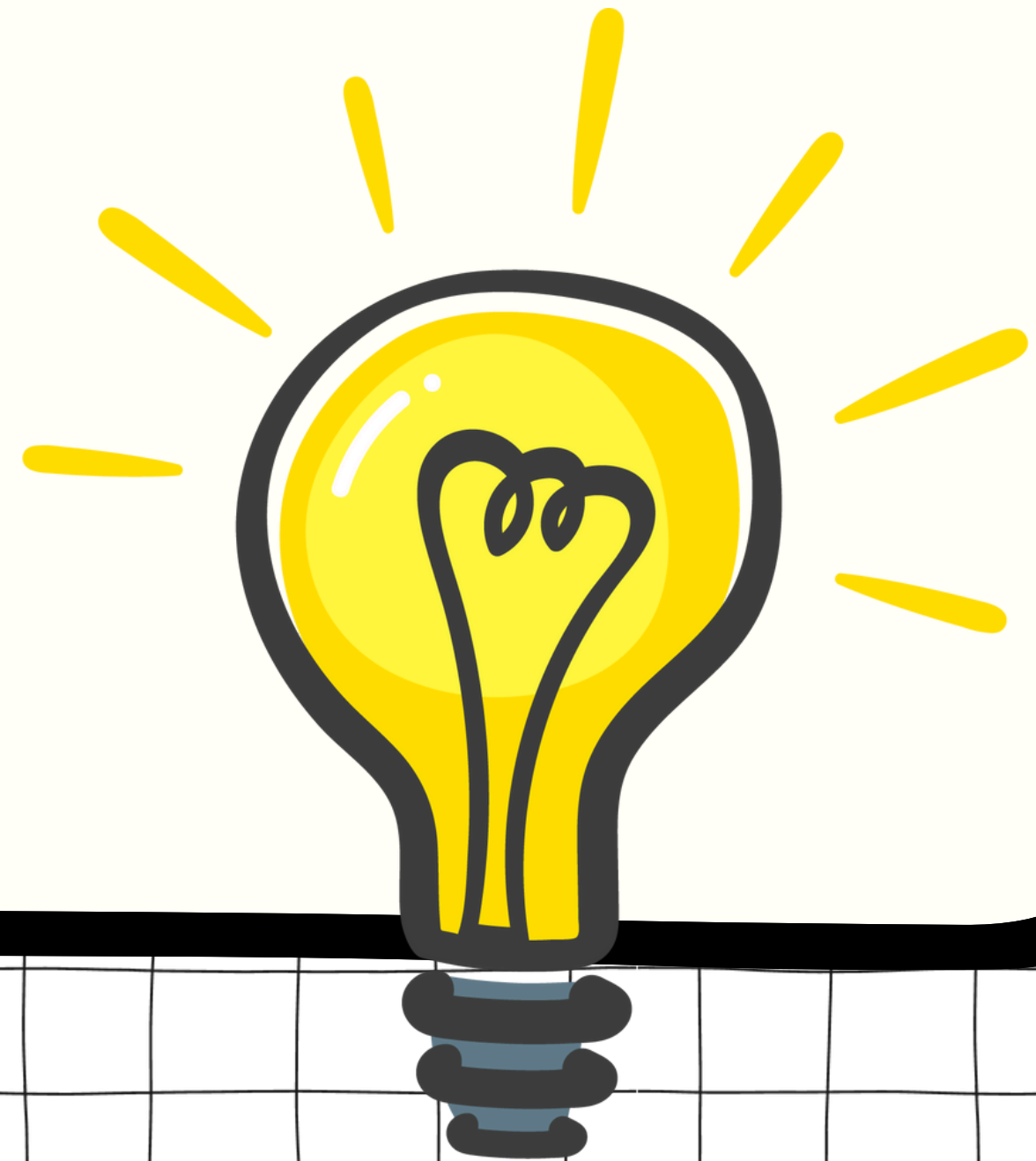
Team – SmartLabs

Name – Sahal Hussain

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Smart Labs

AI-Powered Virtual Labs



Problem Statement

Design and develop innovative, scalable, and technology-driven solutions to enhance the OLABs platform, addressing key challenges in virtual education. Focus on improving the accessibility, functionality, and effectiveness of virtual labs while ensuring security, reliability, and data integrity.

Proposed Solution

The proposed solution enhances OLABs with AI-powered virtual labs using LLMs for interactive guidance and AI tutors. It integrates pre-prompting techniques and domain-specific LLMs to provide accurate explanations and AI-generated questions for assessments. Additionally, improvements have been made to the hands-on coding experience, which was previously lacking in OLABs. The solution also includes a sign language translation feature, making the platform more accessible for diverse learners.

**Pretrained
LLMs**

**Cloud
based**

Personalized

**Integration
via APIs**

DEMO

TRY HERE

smartlabs-olabs.vercel.app

smartlabs.sahalhes.me

ENHANCEMENTS

How it enhances O Labs platform

The solution enhances O Labs by integrating AI-powered coding environments and automated feedback, enabling students to write, test, and receive instant feedback on code. It also utilizes LLMs for interactive guidance and AI-generated questions, improving engagement and personalized learning.

ACCESSIBILITY

just 2 lines



```
<script src="script.js"></script>
<script type="module">
  import { CONFIG } from './config.js';
  import { createChat } from 'https://cdn.jsdelivr.net/npm/@n8n/chat/dist/chat.bundle.es.js';
  createChat({
    webhookUrl: CONFIG.WEBHOOK_URL
  });
</script>
```

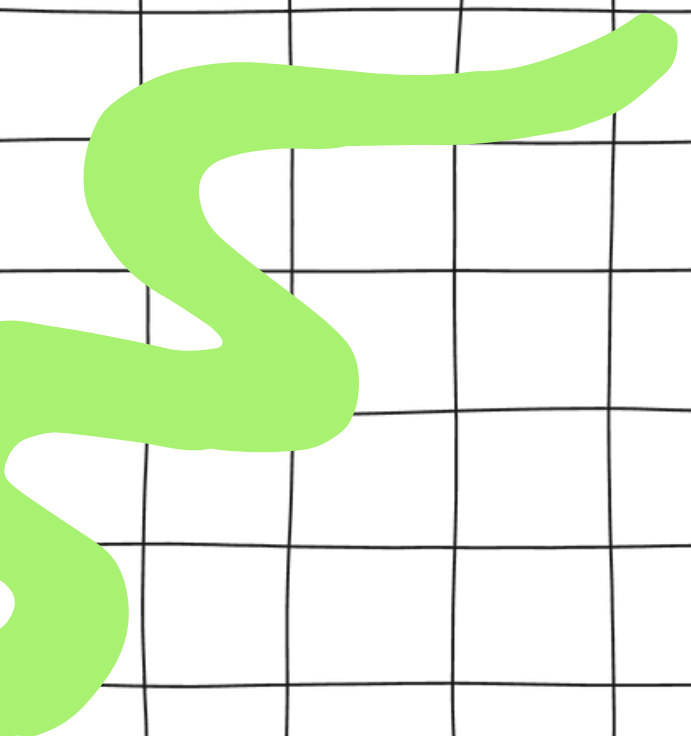
INNOVATION

Enhancing website using AI agents and LLM
Hands on coding with real time feedback
Sign language translation implemented

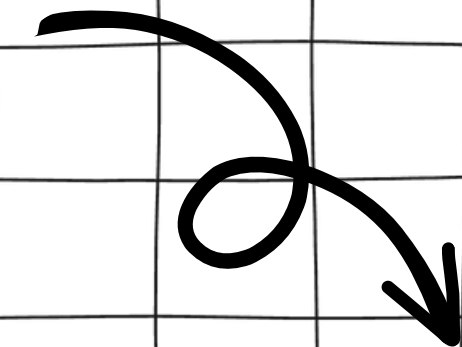
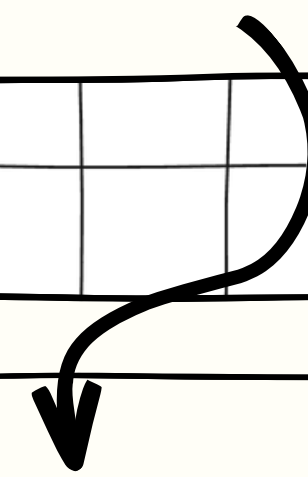
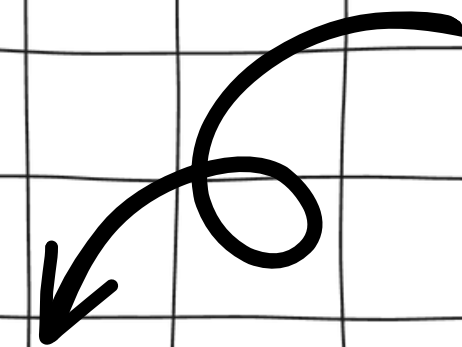
ETHICAL STATEMENT



The project was completely build in hackathon itself
To improve multilingual accessibility, we integrated sign.mt
(<https://github.com/sign/translate>), enabling seamless translations.
This fosters inclusive learning experiences, ensuring students
worldwide can access quality education in their preferred language
while upholding academic integrity and equitable opportunities.



Conclusions

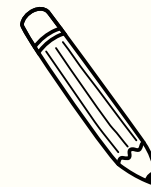


Hands on coding



Our enhancements enable hands-on coding with real-time AI feedback, fostering interactive learning and improving problem-solving skills in O Labs.

Real time feedback



The system provides instant AI feedback, helping students correct mistakes, deepen understanding, and reinforce learning with adaptive hints and suggestions.

Effective



AI-driven virtual labs adjust difficulty, recommend relevant experiments, and provide intelligent assistance, enhancing engagement and making learning more personalized and effective.

